HTML5 –API- Day-6 Hands On – Chintha Ramakrishna

Problem 1 :-

Users can type notes in a textarea and save them in the browser without any server.

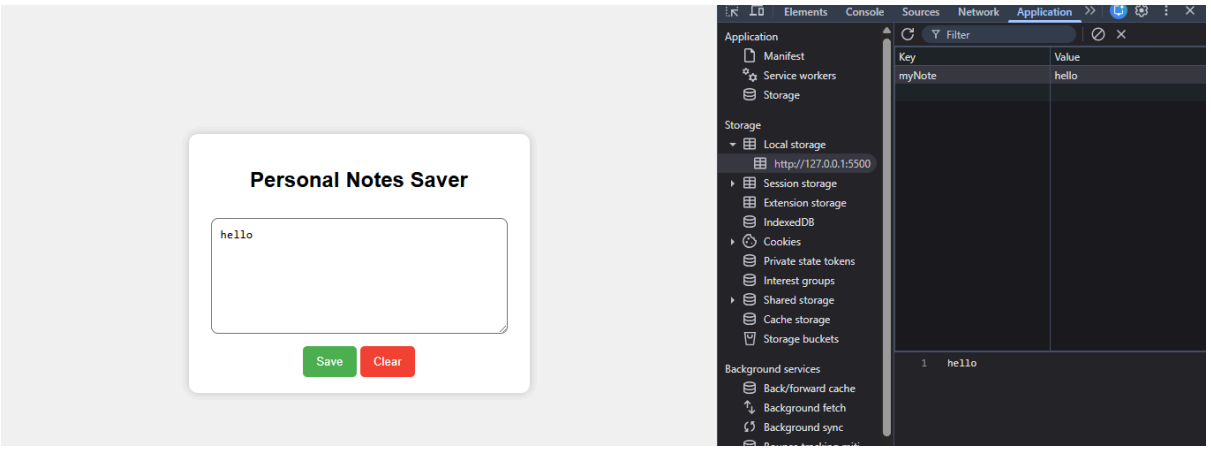
Code :-

```
<!DOCTYPE html>
<html>
<head>
    <title>Personal Notes Saver</title>

    <style>
        /* Basic page styling */
        body{
            font-family: Arial;
            background:#f2f2f2;
            display:flex;
            justify-content:center;
            align-items:center;
            height:100vh;
            margin:0;
        }

        .container{
            background:white;
            padding:18px;
            border-radius:10px;
            width:350px;
            box-shadow:0 0 10px rgba(0,0,0,0.2);
            text-align:center;
        }

        textarea{
            width:100%;
            height:120px;
            padding:10px;
            margin-top:10px;
            border-radius:8px;
        }
```

```
        button{
            margin-top:10px;
            padding:10px 15px;
            border:none;
            border-radius:5px;
            cursor:pointer;
        }

        .save{ background:#4CAF50; color:white; }
        .clear{ background:#f44336; color:white; }
    </style>

    <script>
        /* Load saved note when page loads */
        function loadNote(){
            let savedNote = localStorage.getItem("myNote");   //
getItem()
            if(savedNote){
                document.getElementById("note").value = savedNote;
            }
        }

        /* Save note to localStorage */
        function saveNote(){
            let noteText = document.getElementById("note").value;
            localStorage.setItem("myNote", noteText);   // setItem()
            alert("Note Saved!");
        }

        /* Clear note */
        function clearNote(){
            localStorage.removeItem("myNote");   // removeItem()
            document.getElementById("note").value = "";
        }
    </script>
</head>

<!-- body onload event -->
<body onload="loadNote()">

    <div class="container">
        <h2>Personal Notes Saver</h2>
```

```
        <!-- Requirement: textarea for writing notes -->
        <textarea id="note" placeholder="Write your notes
here..."></textarea>

        <br>

        <!-- Requirement: Save button using inline onclick -->
        <button class="save" onclick="saveNote()">Save</button>

        <!-- Requirement: Clear button -->
        <button class="clear" onclick="clearNote()">Clear</button>

        <!-- Requirements in comments:
            ✔ textarea for notes
            ✔ Save button with onclick
            ✔ Clear button
            ✔ localStorage.setItem()
            ✔ localStorage.getItem()
            ✔ localStorage.removeItem()
            ✔ body onload to display saved note
            ✔ Pure HTML + JavaScript (no backend)
            ✔ Data stored as key-value pair
        -->
    </div>

</body>
</html>
```

Output :-



Explanation :-

1. This project allows users to write daily notes on a web page.
2. When the user clicks Save, the note is stored in the browser using localStorage as a key-value pair.
3. When the page refreshes, the saved note automatically loads and displays in the textarea.
4. The Clear button removes the saved note from localStorage and clears the textarea.
5. This helps understand onclick events, browser storage, page load handling, and DOM manipulation.

Problem 2 :-

Create a Registration Form that validates Name, Email, and Age using onchange events and stores valid data in sessionStorage.

Code :-

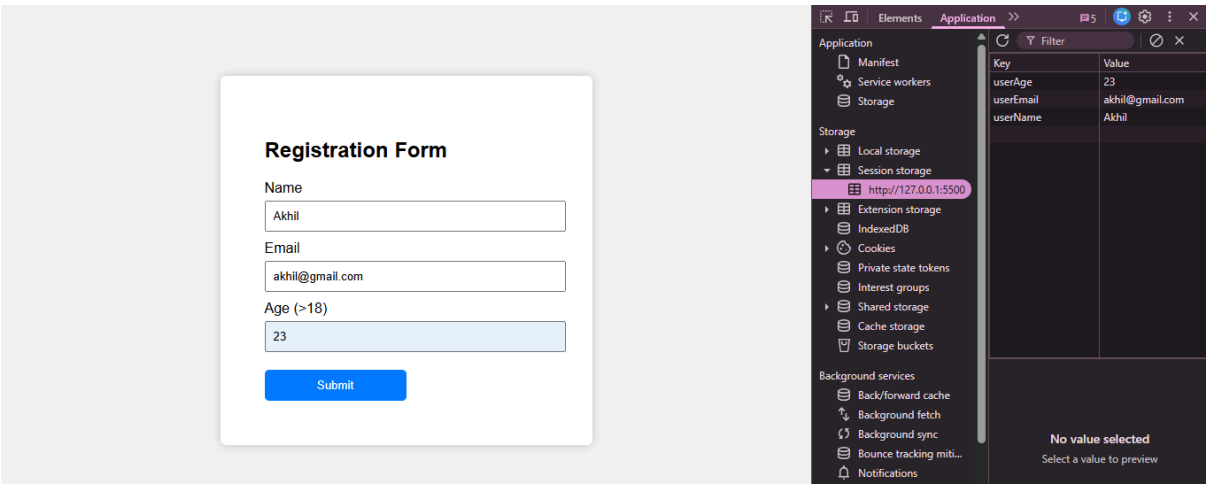
```
<!DOCTYPE html>
<html>
<head>
  <title>Live Form Validation</title>

  <style>
    body{
      font-family: Arial;
      background:#f2f2f2;
    }

    .box{
      width:320px;
      margin:80px auto;
      background:#fff;
      padding:50px;
      border-radius:6px;
      box-shadow:0 0 10px rgba(0,0,0,0.2);
    }
```

```
    label{
      display:block;
      margin-top:10px;
    }

    input{
      width:100%;
      padding:8px;
      margin-top:5px;
    }

    p{
      margin:5px 0;
      font-size:14px;
      color:red;   /* validation color */
    }

    button{
      width:50%;
      padding:10px;
      margin-top:15px;
      background:#007bff;
      color:white;
      border:none;
      border-radius:5px;
      cursor:pointer;
    }
  </style>
</head>

<body>

<div class="box">
<h2>Registration Form</h2>

<label>Name</label>
<input type="text" id="name" onchange="validateName()">
<p id="nameMsg"></p>

<label>Email</label>
<input type="text" id="email" onchange="validateEmail()">
<p id="emailMsg"></p>
```

```
<label>Age (>18)</label>
<input type="number" id="age" onchange="validateAge()">
<p id="ageMsg"></p>

<button onclick="saveData()">Submit</button>
</div>

<script>
function validateName(){
  let name=document.getElementById("name").value;
  document.getElementById("nameMsg").innerText =
    name ? "" : "Name cannot be empty";
}

function validateEmail(){
  let email=document.getElementById("email").value;
  document.getElementById("emailMsg").innerText =
    email.includes("@") ? "" : "Email must contain @";
}

function validateAge(){
  let age=document.getElementById("age").value;
  document.getElementById("ageMsg").innerText =
    age>18 ? "" : "Age must be greater than 18";
}

function saveData(){
  let name=document.getElementById("name").value;
  let email=document.getElementById("email").value;
  let age=document.getElementById("age").value;

  if(name && email.includes("@") && age>18){
    sessionStorage.setItem("userName",name);
    sessionStorage.setItem("userEmail",email);
    sessionStorage.setItem("userAge",age);
    alert("Data stored in sessionStorage");
  }else{
    alert("Please enter valid details");
  }
}
</script>

</body>
```

```
</html>
```

Output :-



Explanation :-

1. This project creates a simple registration form with Name, Email, and Age fields.
2. Validation runs whenever the user changes a field using the onchange event.
3. The Name should not be empty, Email must contain @, and Age must be greater than 18.
4. Validation messages are displayed dynamically on the page.
5. If all inputs are valid, the data is stored in sessionStorage, which lasts only for the browser session

Problem 3 :-

How can we fetch the user's current location using Geolocation API and store the last 5 locations in localStorage?

Code :-

```
<!DOCTYPE html>

<html>
```

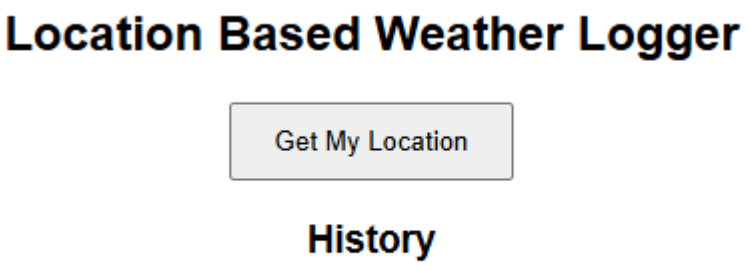
```html
<head>
  <title>Location Logger</title>

  <style>
    body{
      font-family: Arial;
      text-align:center;
      margin-top:50px;
    }
    button{
      padding:10px 20px;
    }
    ul{
      list-style:none;
    }
  </style>
</head>

<body onload="showHistory()">

<h2>Location Based Weather Logger</h2>

<!-- Requirement: Inline event -->
<button onclick="getLocation()">Get My Location</button>
```

```
<p id="current"></p>


<h3>Last 5 Locations</h3>

<ul id="history"></ul>


<script>


function getLocation(){

  if(navigator.geolocation){

    navigator.geolocation.getCurrentPosition(success, error, {

      timeout:5000

    });

  }else{

    alert("Geolocation not supported");

  }

}


/* Requirement: Success callback */

function success(position){

  let lat = position.coords.latitude;

  let lon = position.coords.longitude;


  document.getElementById("current").innerText =

    "Latitude: " + lat + " | Longitude: " + lon;
```

```
  saveLocation(lat, lon);
}

/* Requirement: Error callback */
function error(err){
  let msg="";
  if(err.code==1) msg="Permission denied";
  else if(err.code==2) msg="Location unavailable";
  else if(err.code==3) msg="Timeout";

  alert(msg);
}

/* Requirement: Store JSON and keep last 5 */
function saveLocation(lat, lon){
  let history = JSON.parse(localStorage.getItem("locations")) || [];

  history.unshift({
    latitude:lat,
    longitude:lon,
    time:new Date().toLocaleString()
  });

  history = history.slice(0,5);
```

```
  localStorage.setItem("locations", JSON.stringify(history));


  showHistory();

}


/* Requirement: Display history on load */

function showHistory(){

  let history = JSON.parse(localStorage.getItem("locations")) || [];


  let list=document.getElementById("history");

  list.innerHTML="";


  history.forEach(loc=>{

    let li=document.createElement("li");

    li.innerText = loc.latitude+" , "+loc.longitude+" ("+loc.time+")";

    list.appendChild(li);

  });

}


</script>


</body>

</html>
```

Output :-



Explanations :-

1. The Geolocation API is used to get the user's latitude and longitude with browser permission.

2. The `getCurrentPosition()` method has success and error callbacks to handle results.

3. Location data is stored in localStorage as JSON using `JSON.stringify()`.

4. When the page loads, stored data is read using `JSON.parse()` and displayed as history.

5. Only the last 5 locations are kept by removing older entries.

Problem 4 :-

How can we build a mini expense tracker using a client-side database and SQL queries in the browser?

Code :-

```
<!DOCTYPE html>

<html>
```

```html
<head>
<title>Mini Expense Tracker</title>
<style>
body{font-family:Arial;text-align:center;margin-top:40px}
input{margin:5px;padding:6px}
button{margin:5px;padding:8px}
table{margin:auto;border-collapse:collapse}
td,th{border:1px solid #ccc;padding:6px}
</style>
</head>
<body onload="viewExpenses()">

<h2>Mini Expense Tracker</h2>

<input type="text" id="title" placeholder="Expense Title">
<input type="number" id="amount" placeholder="Amount">
<input type="date" id="date">

<br>

<button onclick="addExpense()">Add Expense</button>
<button onclick="viewExpenses()">View Expenses</button>

<table>
<thead>
```

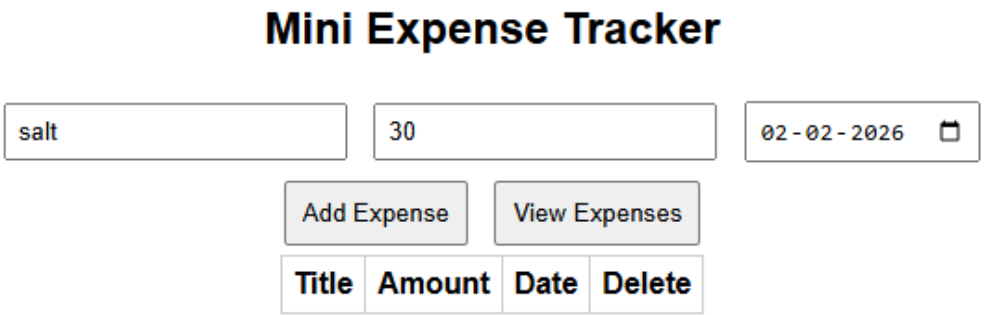
```
<tr>

<th>Title</th>

<th>Amount</th>

<th>Date</th>

<th>Delete</th>

</tr>

</thead>

<tbody id="list"></tbody>

</table>


<script>

let db=openDatabase("ExpenseDB","1.0","Expense DB",2*1024*1024);


/* CREATE TABLE */

db.transaction(function(tx){

tx.executeSql(

"CREATE TABLE IF NOT EXISTS expenses (id INTEGER PRIMARY KEY
AUTOINCREMENT,title TEXT,amount REAL,date TEXT)"

);

});


/* ADD */

function addExpense(){

let title=document.getElementById("title").value;

let amount=document.getElementById("amount").value;
```

```
let date=document.getElementById("date").value;


if(!title || !amount || !date){
alert("Fill all fields");
return;
}


db.transaction(function(tx){
tx.executeSql(
"INSERT INTO expenses (title,amount,date) VALUES (?,?,?)",
[title,amount,date],
function(){
viewExpenses(); // refresh list
},
function(tx,error){
alert("Insert error: "+error.message);
}
);
});


/* clear fields */
document.getElementById("title").value="";
document.getElementById("amount").value="";
document.getElementById("date").value="";
}
```

```
/* VIEW */
function viewExpenses(){
db.transaction(function(tx){
tx.executeSql(
"SELECT * FROM expenses",
[],
function(tx,res){
let tbody=document.getElementById("list");
tbody.innerHTML="";

for(let i=0;i<res.rows.length;i++){
let row=res.rows.item(i);

let tr=document.createElement("tr");

tr.innerHTML=`
<td>${row.title}</td>
<td>${row.amount}</td>
<td>${row.date}</td>
<td><button onclick="deleteExpense(${row.id})">Delete</button></td>
`;

tbody.appendChild(tr);
}
```

```
},
function(tx,error){
alert("Select error");
}
);
});
}

/* DELETE */
function deleteExpense(id){
db.transaction(function(tx){
tx.executeSql(
"DELETE FROM expenses WHERE id=?",
[id],
function(){viewExpenses();},
function(tx,error){alert("Delete error");}
);
});
}
</script>

</body>
</html>
```

Output :-



Explanation :-

1. A client-side database like Web SQL allows storing structured data inside the browser without a server.

2. We create a table using SQL query `CREATE TABLE` inside a transaction.

3. Expenses are added using `INSERT`, viewed using `SELECT`, and removed using `DELETE`.

4. Transactions ensure data safety and help handle errors during query execution.

5. The expense list is dynamically rendered on the page using JavaScript DOM.